**UNIVERSITY OF SAN JOSE-RECOLETOS**
**School of Engineering**
**Computer Engineering Department**

**Evaluating Crowd Management Strategies during Events at Cebu City Sports Center**

A Research Study Submitted to the Faculty of the

Department of Computer Engineering

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**RECOMMENDATION**

PROPOSAL DEFENSE FOR METHODS OF RESEARCH

*This Recommendation is issued on March 17, 2025 with thesis entitled, **AI - POWERED PEOPLE COUNTING WITH REAL - TIME AND HISTORICAL OCCUPANCY ANALYTICS FOR FACILITY MANAGEMENT**, prepared by Eric Mar A. Geonzon, Chashmyr Ignacio, Mykylla Angelyka O. Jordan for oral defense.*

*The manuscript has been fully examined and found to sufficiently fulfill the requirements needed for this course prior to the oral examination.*

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# CHAPTER 1

## INTRODUCTION

### 1.1 Introduction

Mass gatherings which include concerts, religious ceremonies, political protests, and sporting events happen everywhere all around the world. While these events bring people together, they can also create a significant challenge in maintaining public safety and order. Large crowds often carry concerns, especially in cramped areas, such as the possibility of stampedes, overcrowding, and delayed emergency response time. These worries can become even more severe when venues frequently go over their intended capacity, which makes it possible for even minor accidents to turn into serious problems. Implementing effective crowd management is essential to prevent incidents such as stampedes, overcrowding, and other emergencies that can endanger the attendee's safety and can further disrupt the community events.

The Cebu City Sports Center (CCSC), in particular, effectively demonstrates these issues. As the premier venue for major events like the annual Sinulog Festival and national sporting competitions, the CCSC regularly experiences attendance that far exceeds its official capacity of 12,000 seated spectators. The persistent pattern of overcrowding is clearly demonstrated in Table 1, which documents attendance records from recent high-profile events:

**Table 1. Event Attendance and Staffing Records at Cebu City Sports Center**

| Event | Occupancy Limit | Attendees | Deployed Personnel |
|---|---|---|---|
| Arat na Cebu (2022) - "Sinulog is Back" Concert | 20,000 | 25,000 | 250 |
| Sinulog (2024) | 20,000 | 30,000 | Not specified |
| Palarong Pambansa (2024) - Opening Ceremony | Not specified | 41,000 | 80 |
| Sinulog (2025) | 12,000 | 30,000 | Not specified |

Table 1 presents several safety concerns, in the 2022 "Arat na Cebu" concert they exceeded its occupancy limit by 25% while deploying only 250 personnel which means there was one staff member for every 100 attendee's, which is below international safety standards. What is even more alarming is that there were 41,000 guests at the 2024 Palarong Pambansa opening ceremony with just 80 security personnel. Attendee safety is compromised by these tendencies, which show systematic problems in capacity enforcement, staff allocation, and risk assessment.

This study tackles these crowd management problems by carefully examining the real attendance records, how the venue operates, and the existing safety measures. The research aims to achieve three main goals. First, to uncover why overcrowding keeps happening at CCSC events. Second, to compare the current crowd control methods with proven international standards to see where the improvements are needed. Lastly, to suggest practical, data-backed solutions that make events safer while still preserving their cultural importance.

Instead of relying on guesses or unproven opinions, the study focuses on solid attendance numbers and visible operational issues. This approach will give venue managers, event planners, and safety officials clear, useful information they can actually apply to better manage large crowds. The findings will help make popular events safer while keeping what makes them special for the community.

This study is important for all event sites, not only the Cebu City Sports Center. Keeping crowds safe is increasingly difficult but crucial as cities expand and more people attend concerts and festivals. This study provides useful strategies to help protect individuals while maintaining the enjoyment and significance of events.

The findings assist to tackle a problem on how to keep cultural festivals engaging while avoiding harmful congestion. By using data and smart technology, we show how venues can stay safe without losing what makes their events special. These practical solutions can be adapted to any place that is facing similar crowd challenges, making celebrations safer for communities everywhere.

### 1.2 Theoretical Background

Crowd management and risk assessment are critical aspects of ensuring public safety in large gatherings. In order to help the readers understand the algorithms and techniques introduced in this research, the researchers presented theoretical and fundamental outlines on crowd behavior, risk assessment, and technological advancements to provide a foundation for understanding and addressing these challenges.

**Crowd Dynamics Theory** developed by Helbing & Molnár (1995), focuses on the physical movement and flow of crowds in a given space. It examines how groups of people navigate areas, especially in high-density environments, and how obstacles or bottlenecks can disrupt this movement. This theory is essential for crowd management, as it allows for the design of effective crowd flow systems, such as pathways, corridors, and barriers. Predictive models based on crowd dynamics help planners understand potential risks and minimize crowd congestion, which can lead to dangerous situations like stampedes. This theory is applied in transport hubs, event venues, and public spaces to optimize pedestrian circulation and ensure safety during large gatherings.

**Pedestrian Planning and Design theory** (Fruin, 1971) categorizes pedestrian movement into different levels of service (A to F), which reflect the ease with which people can move through spaces. Level A indicates a high degree of freedom, while Level F suggests overcrowded, unsafe conditions. This theory is widely used in the design of urban spaces, particularly when determining the maximum capacity of areas like stadiums, airports, or shopping malls. By assessing pedestrian density and flow rates, planners can ensure that these spaces are designed with enough room to avoid overcrowding and allow for efficient evacuation, helping reduce risk during high-traffic events or emergencies .

**ISO 31000** or **Risk-Based Planning Frameworks** (2018) provide a systematic approach to identifying, assessing, and mitigating potential risks in crowd management. Frameworks like ISO 31000 help planners evaluate hazards related to crowd size, venue capacity, and emergency scenarios. This approach ensures that preventive measures, such as crowd barriers, signage, and emergency protocols, are implemented to minimize risk. Through regular risk assessments and scenario planning, authorities can prepare for emergencies, optimize resource allocation, and ensure that the event venue is safe for attendees. This proactive approach is essential for managing risks in large public gatherings.

**Zoning and Segmentation Strategy** (Bown & Jenkins, 2007) involves dividing large areas into smaller, more manageable zones. Each zone is designed with specific entry and exit points, ensuring that crowd movement is controlled and that people can be easily directed in case of an emergency. This strategy is particularly effective in crowded environments like music festivals or large sporting events, where different zones may require different levels of security, medical support, and crowd control measures. Zoning also allows for staged evacuations, reducing congestion and ensuring that emergency services can access critical areas quickly.

**Ingress and Egress Management** (Rowe & Moore, 2010) refers to the control and optimization of entry and exit points in a venue or event space. This strategy aims to prevent overcrowding at bottleneck areas and ensures that people can enter and exit safely. Effective ingress and egress management includes implementing one-way flow systems, timed ticketing, and smart turnstiles to regulate the flow of attendees. By planning and managing these movement points, event organizers can reduce congestion, enhance crowd safety, and improve the overall experience for attendees. This strategy is commonly applied in concerts, festivals, and stadiums to prevent dangerous crowd situations.

By integrating these theoretical perspectives, this study aims to develop a framework for real-time crowd monitoring using density-based analytics and predictive risk assessment. Understanding crowd behavior and utilizing advanced technological solutions will contribute to improved safety measures and more effective event management strategies.

## 1.3. Literature Review

### Mass Gatherings

Mass gatherings, like sporting events or religious pilgrimages, are highly visible events attended by tens of thousands of people (World Health Organization, 2019). These events pose complex challenges related to public safety, crowd control, emergency preparedness, and health surveillance. The convergence of large populations in limited spaces increases the risk of incidents such as stampedes, heat-related illnesses, disease transmission, and logistical disruptions. According to recent studies, effective planning and management strategies—including risk assessment, crowd behavior analysis, and integration of technology—are essential to prevent accidents and ensure the safety of participants (Algaissi et al., 2021; Darsena et al., 2022). Among these concerns, overcrowding remains one of the most critical and recurrent issues, as it amplifies the potential for disaster by reducing maneuverability, slowing emergency response, and increasing psychological stress within the crowd.

**OverCrowding**

Overcrowding in mass gatherings presents significant safety challenges, particularly in settings where large numbers of individuals converge within confined spaces. As crowd density increases, so does the likelihood of hazards such as stampedes, asphyxiation, delayed emergency response, and difficulty in individual mobility (Khan et al., 2023). These risks are exacerbated when event organizers lack sufficient infrastructure or real-time monitoring systems to manage surges in attendance. Overcrowding also complicates evacuation procedures and increases the strain on medical and security services, reducing their ability to respond efficiently in emergencies. In past events such as the Kumbh Mela and Hajj pilgrimage, overcrowding has led to tragic outcomes, underscoring the need for proactive planning and technology-enhanced monitoring (Zhou et al., 2021). Addressing overcrowding remains a central concern in crowd management, as it directly impacts the safety and well-being of all attendees.

Effective crowd management is crucial for large events , and recent events in the Philippines have shown. The “Sinulog is Back” free concert at the Cebu City Sports

Center (CCSC) on January 16, 2025,  drew more than 30,000 people with another 5,000 to 7,000 turned away due to overcrowding (Magsumbol & Mascardo, 2025).

Subsequently, a concert at Cebu attracted around 100,000 attendees, police estimate said. This event highlighted safety and public health concerns amid the challenge in managing a large number of people. The Cebu City Emergency Operations Center (EOC) has issued guidelines for future crowd-attracting events to guarantee safety and prevent potential health threats, such as an increase in cases of disease following large crowds. These steps emphasize the need for meticulous planning and coordination with the local authorities to effectively handle large crowds (Morexette & Pegeen, 2022).

In Bacolod, it was estimated that 30,000 members of Team Asenso attended a political gathering at the Paglaum Sports Complex. Although the gathering went on smoothly without any reported incidents, the huge crowd highlighted the importance of thorough planning and crowd management techniques to ensure order and safety (Pedrosa, 2024).

This incident clearly brings out the tragic consequences of poor crowd controlling measures and draws attention to better planning and safety arrangements made for such large public events. Managing a huge crowd in closed spaces requires attention to detail with considerations extending beyond mere headcounts. According to Arefin et al. (2024), precise mass counting is essential and involves the adoption of sophisticated technologies, such as density based counting by video analytics and sensor networks to measure the density of people in real time. The physical layout of space should be carefully examined with regard to the barrier's strength, corridors width, the number of emergency exits, and the location to ascertain whether the structure can withstand the expected size of the crowd and evacuate in case of necessity (Alhawsawi et al.). According to Arefin et al. (2024), event managers should adopt an integrated, data-informed process of crowd management. This involves installing systems that monitor crowds in real time with an aid of density-based counting and predictive analytics to forecast areas that may easily lead to overcrowding. The structures of the venues should be analyzed stringently and transformed if necessary to have enough capacity, safe flows, and multiple accessible emergency exits. Besides that, effective trained staff with communication guidelines play a role in guiding and assisting attendees, especially during critical moments (Alhawsawi et al., 2024).

**People Counter in Businesses and Public Institutions**

Private businesses and public institutions like corporate offices, event venues, hospitality, healthcare, libraries, museums, transportation, even worship places (Lepine, 2023) are adopting people counting systems to track and count visitors. Businesses these days rely more on data to improve and expand. Automated people counters' ability to provide accurate and time-related visitor traffic data plays a key role in making better-informed decisions about everything from staffing and marketing to customer experience, profitability, financing, and even people's safety (Lepine, 2023). Automated counters are paving a way for businesses to optimize and grow their services and operations. Thus, by analyzing foot traffic data, businesses can understand which promotions or events attract more customers (Ordedara, 2024).

**Theoretical Foundations of Crowd Management**

Crowd management theories provide a foundation for understanding crowd behavior and designing effective strategies for large gatherings. Crowd Dynamics Theory, introduced by Helbing and Molnár (1995), models pedestrian flow in high-density environments, focusing on how obstacles and bottlenecks affect movement. This theory is widely applied in the design of public infrastructures like transit stations and stadiums to predict flow patterns and reduce the risk of stampedes (Helbing et al., 2000). Similarly, Pedestrian Planning and Design Theory (Fruin, 1971) introduces the concept of pedestrian Level of Service (LOS), which ranges from A (free movement) to F (severe congestion). This theory helps in the design of urban spaces and event venues to avoid overcrowding and ensure safe crowd dispersal (Fruin, 1987). The ISO 31000 Risk-Based Planning Framework (2018) provides a structured approach to identifying, analyzing, and mitigating risks in crowded environments, emphasizing the need for preventive measures such as barriers, signage, and emergency protocols. The Zoning and Segmentation Strategy (Bown & Jenkins, 2007) divides large venues into smaller, manageable zones with specific entry/exit points, allowing for better monitoring and localized interventions. This strategy is particularly effective in large events like festivals, where crowd density varies across zones (Jenkins & Brown, 2007). Finally, Ingress and Egress Management (Rowe & Moore, 2010) focuses on optimizing entry and exit points to reduce congestion at chokepoints, employing measures such as timed admissions and smart access systems. This approach has been successfully applied in large-scale sports events and concerts to ensure smooth crowd movement and minimize evacuation delays.

## Factors and Indicators of Overcrowding

One of the primary factors contributing to overcrowding is high crowd density and space constraints. Overcrowding occurs when the number of individuals in a given area exceeds the safe capacity, leading to restricted movement and increased risks of accidents. Fruin (1993) defines critical crowd density as exceeding 4-5 persons per square meter, at which point individuals lose voluntary movement, and dangerous crowd turbulence can occur. A real-world example is the 2015 Hajj stampede in Mina, Saudi Arabia, where extreme crowd density and blocked movement paths led to over 2,000 fatalities (Helbing et al., 2007).

Another major factor is poor crowd flow and bottlenecks, which occur when people are unevenly distributed due to narrow pathways, improper entry/exit design, or obstacles blocking movement. According to Helbing and Johansson (2010), crowd disasters often happen when flow rate decreases to less than 0.5 persons per second per meter, causing dangerous pressure buildup. A tragic example is the 2010 Love Parade Disaster in Germany, where a bottleneck at a tunnel entrance caused panic, leading to a fatal stampede (Helbing & Mukerji, 2012).

## Occupancy-Ratio Model as Data Interpretation

The occupancy-ratio model is a quantitative framework used to assess overcrowding by measuring the relationship between available space and the number of occupants, movement flow, and congestion points. Several data models can be applied to interpret these factors effectively, ensuring real-time monitoring, prediction of congestion, and crowd safety management.

**Crowd Occupancy**

The Occupant Density Model (ODM) is commonly used to determine safe occupancy levels by calculating pedestrian density per square meter (Fruin, 1993). This model establishes that safe pedestrian density is below 2 persons/m², while densities exceeding 4-5 persons/m² indicate high risk of overcrowding, movement restrictions, and potential stampedes (Helbing et al., 2000).

**Crowd Movement Flow and Congestion Hotspot**

The Pedestrian Flow Model (PFM), developed by Helbing and Johansson (2010), describes how individuals move based on speed, density, and directional forces. This model utilizes the Fundamental Diagram of Pedestrian Flow, which states that flow rate (persons per second per meter) decreases as density increases, leading to slower movement and bottlenecks (Seyfried et al., 2006). The Bottleneck Model, developed by Helbing et al. (2000), predicts congestion by analyzing how narrow pathways, obstacles, and entry/exit points influence pedestrian buildup. This model identifies critical areas where the flow rate falls below 0.5 persons/second/meter, increasing the probability of dangerous pressure accumulation.

**Crowd Management Strategies**

**Physical Infrastructure Design**

The use of physical infrastructure such as barriers, fencing, gates, and clearly designated pathways help guide crowd movement, reduce confusion, and prevent overcrowding in specific areas. Properly designed barriers can also mitigate risks of stampedes and allow emergency responders clear access during incidents. According to Still (2014), infrastructure design plays a critical role in controlling crowd dynamics, especially in high-risk environments such as religious pilgrimages and festivals, where pedestrian flow must be highly regulated to prevent tragedy.

Figure #. Hajj pilgrimage in Mecca, barriers and pedestrian bridges have been constructed to control movement and prevent deadly stampedes (Newswires, 2024)

**Zoning and Segmentation**

This method involves dividing a venue or area into smaller, more manageable sections to control crowd density and streamline movement. Each zone can be monitored independently, allowing authorities to respond quickly if problems arise in a specific segment. This strategy enhances overall safety and helps manage large-scale events by simplifying crowd oversight and reducing the risk of mass panic or congestion. Bown and Jenkins (2007) emphasize that zoning is especially effective when combined with color-coded signage and designated staff per zone to maintain order and assist with directions.

## Ingress and Egress Planning

Effective planning of ingress and egress, the controlled entry and exit of attendees, is essential to avoid congestion and ensure smooth movement, particularly in emergencies. Organizers use techniques like timed entry, staggered access points, and one-way routes to manage the flow of people. Poor ingress and egress planning has been linked to several crowd disasters in history. Rowe and Moore (2010) argue that pre-event modeling of crowd flow and clearly marked routes significantly improves the safety and comfort of attendees while also reducing bottlenecks at critical points.

### Real-Time Monitoring and Surveillance

Advanced technologies such as closed-circuit television (CCTV), drones, and thermal imaging are now widely used to monitor crowd behavior in real time. Surveillance systems can detect rising crowd density, unusual movements, or potential safety threats, allowing organizers to respond proactively. Dandekar and Patnaik (2015) highlight how integrating Internet of Things (IoT) sensors with video analytics enables continuous crowd assessment, which is particularly useful in large open-air events where human monitoring alone may be insufficient.

### Crowd Communication Systems

Communication is a critical aspect of managing large groups of people effectively. Public address systems, digital signage, and mobile alerts inform attendees about schedules, delays, or emergencies, helping reduce uncertainty and panic. Clear communication ensures that the crowd follows designated instructions, especially during evacuations or rerouting. According to Haghani and Sarvi (2017), well-timed and clearly delivered instructions significantly influence how people respond during high-stress situations, ultimately determining the success of crowd dispersal efforts.

### Risk Assessment and Contingency Planning

Risk-based planning is essential for identifying potential hazards and preparing appropriate responses. This includes conducting hazard assessments, establishing safety protocols, and running emergency drills prior to an event. Frameworks like ISO 31000:2018 provide standardized guidelines for identifying and managing risks in various event settings. Implementing a thorough risk management process helps prevent incidents and equips staff with contingency strategies for worst-case scenarios such as fires, medical emergencies, or terrorist threats (ISO, 2018).

### Training and Deployment of Crowd Stewards

Trained personnel such as stewards and marshals are integral to crowd management efforts. These individuals are positioned strategically throughout an event venue to provide assistance, maintain order, and respond to emerging risks. According to the UK’s Health and Safety Executive (2000), properly trained staff can de-escalate tense situations, guide evacuations, and offer real-time support, especially when crowd behavior becomes unpredictable. Their presence also reassures attendees and reinforces a sense of organization and control.

### Measurement Tools for Crowd Satisfaction

Post-event surveys are among the most effective tools for evaluating attendee satisfaction, particularly regarding crowd management, safety, and the overall event experience. Al-Qurashi et al. (2023) highlight the importance of structured questionnaires in evaluating crowd management quality at large-scale events. In their study during the Riyadh Season, they used survey items that addressed accessibility, security visibility, signage clarity, queuing efficiency, emergency preparedness, and health safety protocols. Questions were formatted using a 5-point Likert scale, allowing for consistent quantification of subjective experiences.

Event Impacts (2025) recommends including both quantitative and qualitative questions in post-event surveys. Their suggested framework includes overall satisfaction, specific crowd management elements, emotional responses, and perceived efficiency of Open-ended feedback. Cheng et al. (2021) emphasize the psychological dimension of crowding, noting that survey questions should assess perceived crowding (spatial and social), not just physical density. ModernMeetingStandard.com (2025) suggests including logistical satisfaction indicators, which are indirectly tied to crowd control.

**Data Interpretation for Crowd Satisfaction**

Descriptive statistics such as mean, standard deviation, frequency, and percentage are commonly used to analyze responses collected from Likert-scale instruments. These tools allow researchers to measure the central tendency and variability of participant feedback, offering a snapshot of overall satisfaction levels across key aspects of crowd management (Daugherty & Wilson, 2021). Liu et al. (2022) emphasize the effectiveness of the five-point Likert scale in such assessments, particularly in evaluating service quality, flow of crowd movement, safety, and information dissemination during events. In this format, each response is assigned a numerical value, enabling the computation of average scores for each item and dimension. Cronbach's Alpha is frequently used to measure the internal consistency of grouped items. According to Reyes and Tan (2023), a Cronbach's Alpha coefficient of 0.70 or higher indicates acceptable reliability, while a score above 0.80 demonstrates strong consistency among survey items targeting the same construct. This is especially relevant when analyzing grouped questions related to crowd navigation, safety, or emergency preparedness. In addition to quantitative methods, qualitative responses from open-ended questions are increasingly being integrated into post-event evaluations. Gómez and Choi (2020) highlight that thematic analysis is a widely used method for organizing and interpreting textual data.

## Technological Interventions for Crowd Management

### AI-Powered Video Analytics

Artificial Intelligence (AI)-driven video analytics have revolutionized crowd management by enabling real-time monitoring and analysis of crowd behaviors. These systems utilize machine learning algorithms to detect anomalies such as sudden crowd surges, unauthorized access, or potential threats. For instance, a study by Adithya et al. (2023) demonstrated the use of AI and machine learning technologies to analyze video feeds from existing CCTV networks, facilitating the identification and assessment of crowd dynamics, early detection of potential criminal activities, and continuous monitoring of workplace environments. This approach enhances public safety measures and improves organizational productivity by leveraging intelligent video analytics without the need for extensive system overhauls

### Geofencing and Location-Based Alerts

Geofencing technology leverages GPS and RFID to create virtual boundaries, triggering alerts when individuals enter or exit designated areas. This technology has been instrumental in managing crowd movements and ensuring safety during large events. For example, Abbas et al. (2019) presented a GPS-based location monitoring system with geofencing capabilities to prevent vehicle theft. The system monitors and tracks vehicle locations, issuing alerts when vehicles exit predefined geofenced areas, demonstrating the potential of geofencing in enhancing security and management in various contexts.

### Crowd Heatmaps and IoT Sensors

The integration of Internet of Things (IoT) sensors in crowd management allows for the real-time tracking of individuals' movements, generating heatmaps that visualize crowd density across different zones. These heatmaps enable authorities to identify potential bottlenecks and redistribute crowds proactively. Kucuk et al. (2020) designed a crowd sensing-aware disaster framework using IoT technologies, which facilitates real-time monitoring and management of crowds during disaster scenarios. The framework's effectiveness in enhancing situational awareness and response strategies underscores the potential of IoT-based crowd heatmaps in improving public safety.

### Mobile Event Safety Apps

Mobile applications have emerged as vital tools in disseminating real-time information and ensuring safety during mass gatherings. These apps provide features such as live updates, emergency alerts, and navigation assistance, enhancing communication between organizers and attendees. Darsena et al. (2022) reviewed sensing technologies for crowd management in public transportation systems, highlighting the role of mobile applications in informing users of crowding status in real-time. The integration of such apps contributes to improved user comfort and satisfaction during normal operations and enhances the ability to cope with emergency situations.

**Digital Twin and Simulation Modeling**

Digital twin technology involves creating virtual replicas of physical environments, enabling planners to simulate various scenarios and optimize crowd management strategies. By modeling pedestrian flow and testing emergency evacuation plans, authorities can identify potential issues and implement solutions proactively. Zhong et al. (2024) proposed a smart control scheme for airport crowds based on digital twin technology, constructing an integrated crowd control system framework that improves the efficiency and intelligence level of airport crowd control, providing technical support for the construction of intelligent airports.

**Facial Recognition for Crowd Screening**

Facial recognition technology has been increasingly adopted for crowd screening, facilitating the identification of individuals and enhancing security measures. During the COVID-19 pandemic, the Transportation Security Administration (TSA) initiated a pilot program for facial recognition checkpoints at airports to reduce physical interactions between employees and passengers. This technology aims to enhance safety while providing significant security benefits, though it raises privacy concerns that necessitate careful consideration and transparency.

# THE PROBLEM

### 1.4. Statement of the Problem

The researchers conducted this study to determine the factors and characteristics that influenced overcrowding and establish innovative and technological solutions to mitigate and assist crowd management.

Specifically, this study sought to answer the following questions:

1. What are the current strategies, techniques, and technologies used by the Cebu City Sport Complex officials for crowd management?
2. With the current crowd management strategies and techniques, how effective is it in terms of crowd occupancy and attendees' satisfaction?
3. Based on the findings of the study, what strategies and technological interventions can be proposed to enhance crowd management and emergency readiness in public spaces?

### 1.5. Scope and Limitations

This study focuses on the current strategies, techniques, and technologies used for crowd management at the Cebu City Sports Complex. The research specifically examines the effectiveness of these strategies in managing crowd occupancy and ensuring attendees' satisfaction. The primary data collection for this study is based on surveys distributed to event attendees and structured interviews with key personnel such as security staff, event organizers, and facility managers. The study also reviews secondary data, including official documents, crowd control reports, and relevant records from the Cebu City Sports Complex. Furthermore, the research applies analytical models such as the Occupancy Density Model, Cronbach’s Alpha, and Thematic Analysis to interpret the data. The proposed strategies and technological solutions aim to enhance crowd management and emergency preparedness in public spaces based on the findings.

This study is limited to the context of the Cebu City Sports Complex and does not account for crowd management strategies used in other public spaces or cities, which may have different infrastructure or crowd dynamics. The research also relies on self-reported data from surveys, which may be influenced by attendee perceptions and biases. Additionally, the interviews were conducted with a select group of personnel, meaning the findings may not fully represent all stakeholders involved in crowd management at the facility. The absence of on-site observations and real-time monitoring further limits the ability to assess dynamic crowd behavior. The study is also constrained by the availability and completeness of secondary data, such as crowd control reports, which may not cover all past events or emergency situations comprehensively. Finally, technological interventions proposed in this study are based on current trends and may not account for future advancements in crowd management technologies.

### 1.6. Significance of the Study

The current study is crucial as the increased risk of mass gathering overcrowding poses a critical challenge, with potential innovative solutions in effective crowd management strategies. This research aims to address these critical issues by exploring crowd management strategies and formulating technological interventions to enhance crowd management.

This study is going to be beneficial to the following:

**Event organizers and venue managers:** This study will provide them with actionable strategies and technological insights to proactively manage crowds, enhance safety, and optimize the flow of people within their venues (Namreen & Bindhu, 2023).

**Security Personnel and First responders:** The research output will better equip them with better tools and knowledge for real time monitoring, risk assessment and emergency response at mass gatherings that would improve the safety of people in their hands (Namreen & Bindhu, 2023).

**Public Safety agencies and Policymakers:** In this study, the development of more effective regulation, guidelines and training programs on crowd management will be enhanced which contribute to safer public spaces and reduce risk in mass gatherings (Kadi et al., 2024).

**Business and Retailers**: by adopting effective crowd management strategies that are informed by this research, businesses can enhance customer experience, reduce wait times, optimize facility layouts and improve overall operational efficiency (Al-Qurashi et al., 2023).

**Attendees and General Public:** Ultimately, this research aims to improve the safety and well being of individuals attending mass gathering, creating more secure and enjoyable experiences for everyone (Papalexi & Bamford, 2020).

By promoting safer crowd management practices, the study contributes to a more secure public environment. The integration of theoretical perspectives will be social identity theory and self organization theory, with a view to constructing a framework based on real time crowd monitoring involving density based analytics and predictive risk assessment. Crowd behavior, concerning how individuals adapt to shared identity and how a crowd spontaneously acquires movement pattern is also essential for any effective management action.

# FLOW OF RESEARCH

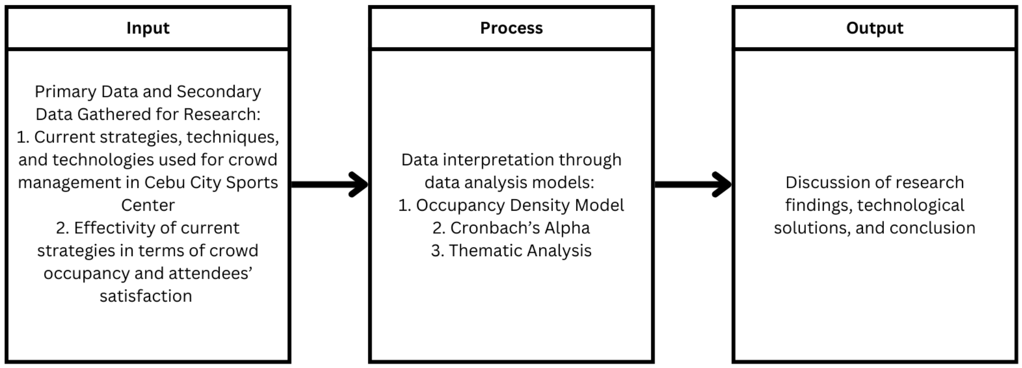


**Figure 2. Research Flow Diagram**

The research began by gathering both primary and secondary data to evaluate the current strategies, techniques, and technologies used for crowd management at the Cebu City Sports Center. Primary data collection involved surveys distributed to event attendees to assess their level of satisfaction with crowd management efforts, as well as structured interviews with key personnel such as security staff and event organizers to gain insights into the effectiveness of existing strategies. Secondary data included the review of official documents, reports, and records related to crowd occupancy and control measures. The collected data was analyzed using the Occupancy Density Model to assess the efficiency of space utilization during events, Cronbach's Alpha to determine the reliability of survey instruments, and Thematic Analysis to extract patterns and key themes from qualitative responses. This process led to a comprehensive discussion of the research findings, the identification of potential technological solutions for enhanced crowd management, and a conclusion that evaluates the effectiveness of current practices and suggests areas for improvement.

# RESEARCH METHODOLOGY

### 1.7. Research Design

The research design of this study follows a qualitative & quantitative approach, implementing both structured interviews and observation for gathering data. By using these techniques, the research hopes to offer a thorough grasp of people counting systems and crowd control techniques. In order to evaluate crowd occupancy rates, movement patterns, and congestion spots, observations will be made in a few high-traffic areas. In addition, planned interviews with important staff members in charge of crowd monitoring will provide insight about the procedures, difficulties, and advancements in people counting methods. To further support the conclusions and offer a more comprehensive assessment of its effectiveness, existing crowd management data will also be examined.

### 1.8. Research Environment

The research environment is the Cebu CIty Sports Complex (CCSC) formerly known as the Abellana Sport Complex is a public, multi-purpose sporting facility located on Osmeña Boulevard in Cebu City, Philippines. The CCSC has become a high-traffic area due to its central location and accessibility to visitors, and It’s regularly overcrowded, especially during big events. The stadium can accommodate up to 16,000 audiences and has a seating capacity of 8,000 people encircling the field. The main stadium features a 398-meter rubberized track falling 0.88 meters short of the standard 400 meters. However, overcrowding has been a persistent problem at the site.

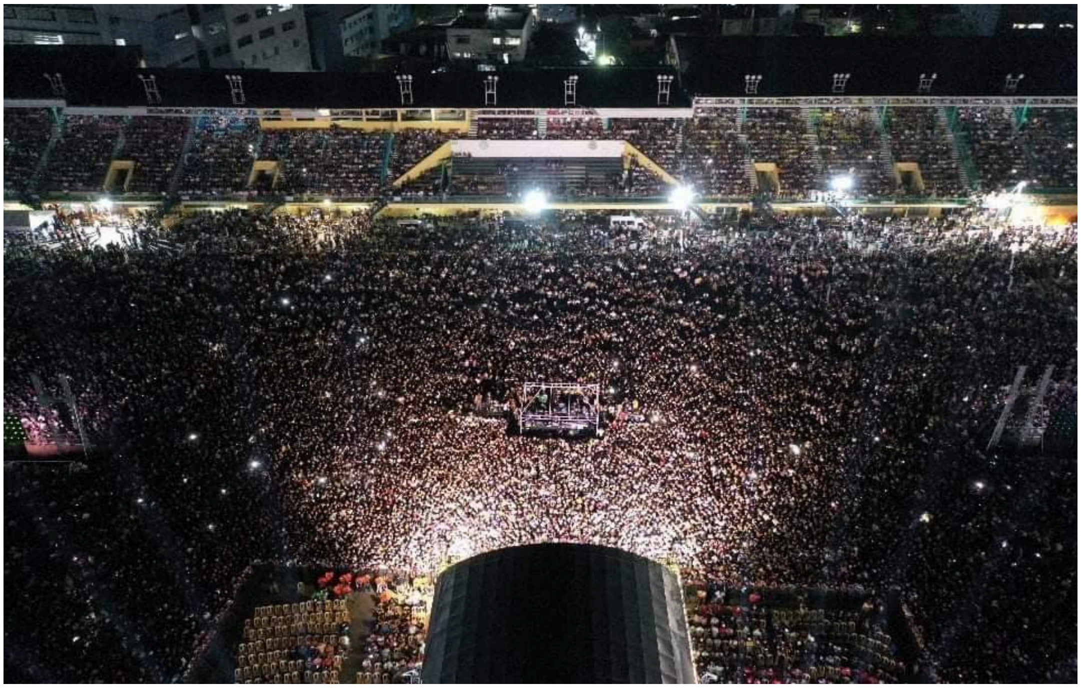

Figure # 3 Drone shot of “Arat na Cebu” Free Concert

Figure 3 shows the free concert "Arat na Cebu," which took place on January 16, 2023, allegedly drew over 100,000 people, much in excess of the venue's intended capacity (SunStar Cebu, 2023).

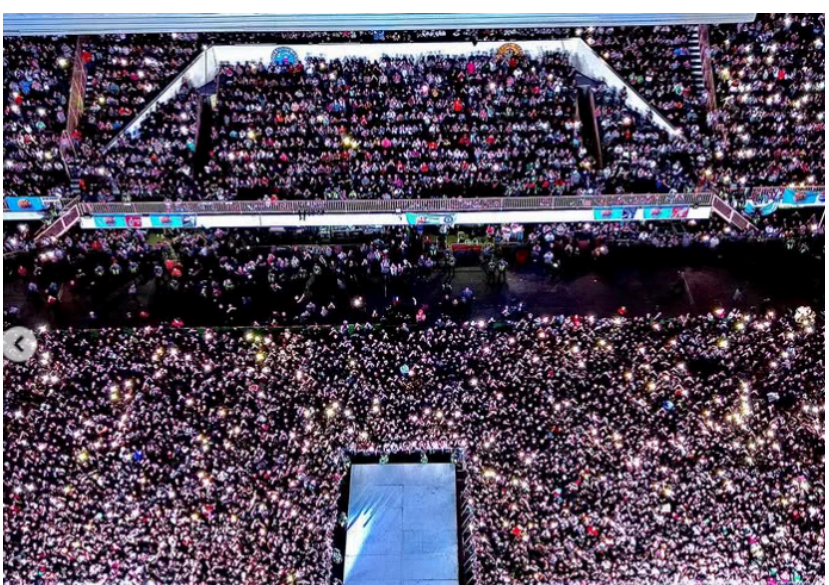

Figure # 4 Inside of the CCSC during “Sinulog is back Concert”

Figure 4 shows the "Sinulog is Back" event, which sparked worries about emergency response, crowd safety, and monitoring restrictions. The event took place within the CCSC and drew over 30,000 attendees.

It is regularly hosted by local, regional, and national sporting events, which includes the annual Sinulog Festival. It is the perfect setting for evaluating and improving crowd control techniques during major public events because of its central position and diverse audience.

**1.9. Research Respondents**

This study will involve a representative sample of respondents who have responsibilities with various aspects of event management, venue operations, security, crowd control and emergency response at the CCSC. Apart from that, respondents who take part in events, such as spectators and participants who perform during an event will be involved. Furthermore, volunteers who assist in different ways during the events will also be involved. These stakeholders shall provide valuable information on current crowd management practices, challenges faced and scope of improvement. Through their integration, the researchers aim to develop balanced strategies for enhancing crowd management safety at the venue.

Realistically, it is not possible to interview a large number of staff and visitors directly because of time and resource constraints. Thus, the researchers will carry out surveys with a representative sample of both groups, focusing on those directly engaged in crowd management. To calculate the sample sizes from  personnel and attendees, the researchers used an online calculator at Calculator.net (2008). The calculations are done at 95% confidence level, a 15% margin of error, and population portion of 50%.

The number of personnel and attendees were estimated in terms of averages from event guests' records and staffing reports of the Cebu City Sports Center. Table 1: *Event Attendance and Staffing Records at Cebu City Sports Center* shows the numbers used for estimating the average number of attendees and personnel of different events.

The number of personnel deployed for the events under consideration were 250, 800, and 80, respectively. To find the average number of personnel, these numbers were added and divided by 3: Average number of personnel = (250 + 800 + 80) / 3 = 376.67. Hence, the number of personnel population was estimated at 377, with a 95% confidence level, 15% margin of error, and 50% population proportion, the calculator proposed a sample size of approximately 39 personnel.

The attendees' numbers for the events in consideration were 100,000, 30,000, and 41,000, respectively. To calculate the average number of attendees, these figures were added and divided by 3: Average number of attendees = (100,000 + 30,000 + 41,000) / 3 = 57,000. Thus, the population size of attendees was estimated to be 57,000. With a confidence level of 95%, a margin of error of 15%, and a population proportion of 50%, the sample size calculator recommended a sample of approximately 43 participants.

Therefore, to make the data more manageable, the researchers will aim to survey around 39 personnel and 43 attendees focusing on those who handle crowd management directly. It is important to acknowledge that real-world limitations may prohibit researchers from attaining the optimal sample size. Nevertheless, this methodology will strike a balance between the requirement for a representative sample and constraints of available time and resources. Data derived from the study will be used to inform the formulation of recommendations to improve crowd management safety and efficiency.

### 1.9. Research Instruments

This research will utilize two primary research instruments: guided interviews and surveys. Guided interviews will be carried out with crowd management personnel to ensure a more detailed analysis of the existing crowd management systems and problems. The interviews will enable a comprehensive understanding of the complexities involved in crowd management, allowing for identifying the key issues and areas for improvement.

The surveys, on the other hand, will be conducted with the attendees to obtain qualitative data regarding their experiences with crowd management. This information will be used to identify patterns and trends in the effects of crowd management practices on attendees. Combined, the qualitative data from interviews and quantitative data from surveys will provide a balanced picture of the issues and areas of improvement in crowd management practices. The expected outcome of these tools is to obtain thorough data that will guide the formulation of efficient and effective measures for improving crowd management safety and efficiency.

### 1.10. Dry-Run Procedure

The researchers will conduct preliminary observations at Cebu City Sports Complex to gather initial insights into crowd dynamics and attendees experiences. Short guided interviews with crowd management staff will also be conducted by the researchers to refine the questions for the main interviews. This initial data collection will act as reference to provide assurance that the questions asked during the main interviews are clear, relevant and aligned with the research objectives. These initial steps will enable the researchers to make necessary adjustments to provide an effective and dependable data collection process. The findings obtained through these initial surveys and interviews would also guide the construction of the final survey questionnaire to be circulated among attendees so that it can obtain the most critical information about their experiences with crowd management.

### 1.11. Data Collection

The data collection for this study will mainly be through guided interviews and surveys. Guided interviews with crowd management officials will be done to seek detailed information on their approach to crowd management, how effective their approach has been, their problems and possible solutions to these problems. These interviews will offer qualitative data on crowd management practice and stakeholder's perceptions and will offer a rich understanding of the issues involved.

Surveys will be handed out to attendees at events held at the venue to gather quantitative data about their experiences with crowd management. The survey will allow us to discern trends and patterns in attendees' views about crowd management practices, yielding useful insights into how these practices affect event goers. Through the integration of the qualitative findings from the interviews and the quantitative data from surveys, the study aims to achieve a comprehensive understanding of crowd management opportunities for improvement and issues. The information gathered would be used to provide insights towards effective strategies on improving crowd management efficiency and safety.

**1.12. Data Treatment**

The gathered from interview responses individuals responsible for event planning, crowd control, traffic management, and public safety will be transcribed and analyzed using thematic coding. The researchers will take note and highlight the recurring themes, challenges, and best practices observed from the responses given by the respondents.

The occupancy rate data will be calculated, as a measure to determine the effectivity of the current crowd management strategies and techniques, using the equation:

$$O_r = \frac{Occupants}{Maximum\ Capacity}\ x\ 100$$

The equation will give as the occupancy rate percentage in which will be interpreted with a threshold of 0% to 70% as the Ideal Threshold, 71% to 90% as Safety Concern, 91% and above as Overcrowding.

The data collected from the post-event survey were systematically encoded, cleaned, and analyzed to determine the level of attendee satisfaction with crowd management practices during the event. All survey responses were initially transferred into a digital spreadsheet for processing. Each item in the questionnaire was assigned a numerical code, particularly the Likert scale responses, which were scored as follows: 1 – Strongly Disagree, 2 – Disagree, 3 – Neutral, 4 – Agree, and 5 – Strongly Agree. This coding facilitated the use of descriptive and inferential statistical methods.

Descriptive statistics were then applied to the quantitative data. These included the computation of the mean which was calculated to determine the central tendency or average response per item using the formula:

$$\overline{X} = \frac{\sum X}{n}$$

where **X** represents the mean, X is the total sum of all responses for a specific item, and **n** is the total number of respondents. The standard deviation was computed to measure the dispersion or variability of responses from the mean using the formula:

$$S = \sqrt{\frac{\sum(X - \overline{X})^2}{n - 1}}$$

for each item, which helped determine the average level of satisfaction and the variability of responses. Frequency and percentage distributions were also calculated to identify the response patterns for each question. To ensure internal consistency among items grouped under the same dimension, Cronbach's Alpha was computed using the formula:

$$\alpha = \frac{k}{k - 1}\left(1 - \frac{\sum\sigma_i^2}{\sigma_t^2}\right)$$

where **k** is the number of items, $\sigma_i^2$ is the variance of each item, and $\sigma_t^2$ is the variance of the total scores. A value of 0.70 or higher was considered acceptable, indicating that the grouped items reliably measured the same construct. In addition to quantitative data, qualitative responses from open-ended questions were analyzed using thematic analysis.

Based on the data gathered from both the data collection method, the researchers will identify the key components or factors that cause complications in crowd management. The researchers will use the data to provide an advanced and technological solution to mitigate or prevent the problem from occurring.

## ETHICAL CONSIDERATIONS

The study will adhere to ethical guidelines to ensure the privacy, safety, and well-being of all participants. Informed consent will be obtained from all interview respondents, ensuring they understand the purpose of the study, the nature of their participation, and their right to withdraw at any time without repercussions. Observations will be conducted in public areas without interfering with individuals' activities, and no personally identifiable information will be collected.

Additionally, all data gathered will be stored securely and used strictly for research purposes. Confidentiality will be maintained by anonymizing responses and ensuring that sensitive information is not disclosed. The research will comply with relevant ethical standards to ensure that participants' rights and dignity are respected throughout the study. Throughout the research process, the following ethical considerations will be observed:

**Voluntary Participation**: All respondents will participate willingly, without coercion, and will have the right to refuse or withdraw at any stage.

**Informed Consent**: Participants will be provided with a clear explanation of the study's purpose, methods, and potential risks before giving their consent.

**Confidentiality and Anonymity**: Personal identifiers will be removed, and responses will be anonymized to protect participants' identities.

**Non-Interference**: Observations will be conducted discreetly without disrupting normal activities or operations at the boardwalk.

**Data Security**: All collected data will be securely stored and accessed only by authorized researchers to prevent unauthorized disclosure.

**Ethical Approval**: The research will comply with institutional and ethical review board requirements to ensure adherence to professional and academic ethical standards.

By following these ethical principles, the study will maintain integrity, fairness, and respect for all involved parties.

# DEFINITION OF TERMS

**Crowd Management** – The strategic planning and implementation of measures to ensure the safety and orderly movement of people in mass gatherings.

**Crowd Crush** – A dangerous situation where individuals in a densely packed crowd are pushed together with immense force, often leading to injuries or fatalities.

**People Counting** – The process of estimating or determining the number of individuals in a particular space using various technologies, such as sensors, computer vision, or RFID systems.

**Computer Vision** – A field of artificial intelligence (AI) that enables machines to interpret and analyze visual data from images or videos for tasks like people detection and tracking.

**Machine Learning (ML)** – A branch of AI that enables systems to learn and improve from data without being explicitly programmed, often used for crowd analysis and movement prediction.

**Deep Learning (DL)** – A subset of machine learning that utilizes neural networks with multiple layers to process large datasets and perform tasks such as object detection and density estimation.

**Density-Based Analytics** – A technique used to analyze the distribution and movement of people in a crowded environment to predict risks and improve safety measures.

**Radio Frequency Identification (RFID)** – A technology that uses radio waves to automatically identify and track objects or individuals equipped with RFID tags.

**Infrared (IR) Sensors** – Sensors that detect heat signatures to track human presence and movement, often used for people counting in low-light conditions.

**Optical Flow** – A method in computer vision used to estimate movement by analyzing the changes in pixels between video frames, often applied in crowd monitoring.

**Density-Based Analytics** – A method of measuring crowd size and distribution by analyzing the number of individuals within a specific space.

**Bottleneck Effect** – A situation where movement is slowed or obstructed due to the narrowing of pathways or limited access points.

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# APPENDIX/APPENDICES

## I. Consent form

**Crowd Management Research Waiver Form**

This Waiver Agreement ("Agreement") is entered into and made as of the date signed below by the undersigned participant ("Participant") and the requesting party ("Requester").

Purpose: The Participant is aware that the Requester is collecting personal and professional data for research on crowd management practices at the Abellana Sports Complex.

Voluntary Participation: The Participant understands that participation in the furnishing of personal information is voluntary and that they can refuse to respond to any question at any time.

Type of Information Collected: The Requester can inquire into, but not be limited to, the following information:

- Personal information (e.g., name, role, contact details)
- Practices and strategies for crowd management at the Cebu City Sports Complex
- Experience in crowd occupancy and flow of movements during events
- Technology utilization in monitoring crowds at the venue
- Historical records regarding crowd management incidents or successes at the Abellana Sports Complex

Confidentiality and Protection of Data: The Requester undertakes to maintain the confidentiality of all personal data and use it for the purpose expressed alone. The personal data shall not be transmitted to third parties without the Participant's express permission unless necessary under law.

Release of Liability: The Participant releases any claims against the Requester for using the information furnished, with the exception of cases of proven abuse or breach of confidentiality.

Consent and Signature: By executing this waiver, the Participant confirms they have read and agreed to the conditions of this Agreement and consent to voluntarily submit the requested information.

Participant's Information:

- Name: ________________________________________
- Position/Title (e.g., Event Organizer, Venue Manager, Security Personnel, Spectator): _________________________________

- Contact Information (Email/Phone Number): ____________________
- Signature: _______________________________________
- Date: _________________________________________

Requester's Information:

- Name: _______________________________________
- Organization (if applicable): _______________________
- Signature: _______________________________________
- Date: _________________________________________

## II. Questionnaire

**Personnel Survey Questionnaire**

Statistical Questions:

1. Average Crowd Size During Peak Events:
    - What is the average crowd size you manage during peak events at the Abellana Sports Complex?
2. Occupancy Limit:
    - What is the maximum occupancy limit set for the Abellana Sports Complex during events?
3. Personnel Deployment:
    - On average, how many security personnel are deployed during peak events at the Abellana Sports Complex?

4. Number of Emergency Exits:

    - How many emergency exits are available for attendees to use?

Qualitative Questions:

1. Crowd Management Strategies:
    - What strategies and techniques are currently used to manage crowds during events at the Abellana Sports Complex?
2. Crowd Monitoring Methods:
    - How do you currently monitor crowd sizes and movement during events at the Abellana Sports Complex?
3. Technology Use in Crowd Management:
    - Are any technologies (e.g., sensors, cameras) used to aid in crowd management at the venue? If so, how effective are they?
4. Communication Protocols:
    - What communication protocols are in place to coordinate crowd management efforts among different teams (e.g., security, medical, event staff)?
5. Staff Training:
    - What specific training do personnel receive to prepare them for managing crowds during events?
6. Challenges Encountered:
    - What are the main challenges or difficulties encountered when managing crowds during events at the Abellana Sports Complex?

    - *Follow-up:* What factors contribute to these challenges, and how are they typically addressed?

**Attendees Survey Questionnaire**

Instructions: Please rate your agreement with the following statements regarding your experience at the Cebu City Sports Complex (CCSC) during the event.

Scale:

- 1 = Strongly Disagree
- 2 = Disagree
- 3 = Neutral
- 4 = Agree
- 5 = Strongly Agree

I. Crowd Management & Safety

1. Crowd Size Perception: The crowd size at the event allowed for comfortable movement.
    - 1 2 3 4 5
2. Ease of Navigation: It was easy to find my way around the venue.
    - 1 2 3 4 5
3. Perceived Safety: I felt safe and secure at the event.
    - 1 2 3 4 5
4. Emergency Preparedness: I believe venue personnel were well-prepared to handle emergencies.

- 1 2 3 4 5

5. Visibility of Security: Security personnel were visible and accessible throughout the venue.
    - 1 2 3 4 5
6. Effectiveness of Crowd Management: The crowd management strategies in place were effective in maintaining order and safety.
    - 1 2 3 4 5

II. Customer Satisfaction

1. Overall Comfort: The venue provided a comfortable environment.
    - 1 2 3 4 5
2. Event Enjoyment: I enjoyed my time at the event.
    - 1 2 3 4 5
3. Value for Time: The event was worth the time I spent attending.
    - 1 2 3 4 5
4. Facilities and Amenities: The facilities and amenities (e.g., restrooms, food vendors) were adequate and well-maintained.
    - 1 2 3 4 5
5. Organization: The event was well-organized.
    - 1 2 3 4 5
6. Overall Satisfaction: Overall, I was satisfied with my experience at the event.
    - 1 2 3 4 5

III. Technology & Innovation

1. Awareness of Tech: I noticed technologies (e.g., sensors, cameras) being used for crowd management.
   - 1 2 3 4 5
2. Tech Effectiveness: I felt technology was used effectively.
   - 1 2 3 4 5

IV. Open-Ended Feedback

1. What specific improvements would you suggest to enhance crowd management at the Cebu City Sports Complex?

# CURRICULUM VITAE

**ERIC MAR A. GEONZON**
Minglanilla, Cebu
ericmar.geonzon.22@usjr.edu.ph

**Personal Information**

Age: 20 years old

Gender: Male

Birthdate: July 9, 2004

Birthplace: Minglanilla, Cebu

Status: Single

**EDUCATIONAL BACKGROUND:**

**Tertiary:**

University of San Jose - Recoletos 2022 - Present

**Secondary:**

Don Bosco Technical College - Cebu Inc. 2016 - 2022

**Elementary:**

Sae Young Christian School 2013 - 2016

Pardo Extension Elementary School 2007 - 2013


**CHASHMYR IGNACIO**

Bulacao Pardo, Cebu City

chashmyr.ignacio.22@usjr.edu.ph


**Personal Information**

Age: 21 years old

Gender: Female

Birthdate: February 04,2004

Birthplace: Roxas City

Status: Single

**EDUCATIONAL BACKGROUND:**

**Tertiary:**

| | |
|---|---|
| University of San Jose - Recoletos | 2022- Present |

**Secondary:**

| | |
|---|---|
| Southwestern University PHINMA | 2020 - 2022 |
| Abellana National School | 2016 - 2020 |

**Elementary:**

| | |
|---|---|
| Cebu City Central School | 2015- 2016 |
| Our Lady of Mt. Carmel Learning Center | 2009- 2015 |
| Roxas International Pre-school | 2007 - 2009 |


**MYKYLLA ANGELYKA O. JORDAN**
**Minglanilla, cebu**
mykyllaangelyka.jordan.22@usjr.edu.ph


**Personal Information:**
Age: 22 years old
Gender: Female
Birthdate: February 6, 2003
Birthplace: USA
Status: single

**EDUCATIONAL BACKGROUND:**

**Tertiary:**

| | |
|---|---|
| University of San Jose-Recoletos | 2022 - present |

**Secondary:**

| | |
|---|---|
| Asian College of Technology | 2019 - 2021 |
| Montessori Academy of Southern Cebu Inc. | 2016 - 2019 |
| BD Child Care and Learning Center | 2015 - 2016 |

**Elementary:**

| | |
|---|---|
| BD Child Care and Learning Center | 2013 - 2015 |
| Leaton Academy | 2012 - 2013 |
| St. Cecilia's College | 2010 - 2012 |
| St. Paul College Foundation, Inc. | 2009 - 2010 |